

LLMs and NLP Models in Cryptocurrency Sentiment Analysis: A Comparative Classification Study

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Abstract

Cryptocurrencies are becoming increasingly prominent in financial investments, as more investors diversify their portfolios and individuals are drawn to their ease of use and decentralized financial opportunities. However, this accessibility also brings significant risks and rewards, often influenced by news and the sentiments of crypto investors, known as crypto signals. This paper explores the capabilities of large language models (LLMs) and natural language processing (NLP) models for sentiment analysis of cryptocurrency-related news articles. We fine-tune state-of-the-art models such as GPT-4, BERT, and FinBERT for this specific task, evaluating their performance and comparing their effectiveness in sentiment classification. By leveraging these advanced techniques, we aim to enhance understanding of sentiment dynamics in the cryptocurrency market, providing insights to inform investment decisions and risk management strategies. The outcomes of this comparative study contribute to the broader discourse on applying advanced NLP models to cryptocurrency sentiment analysis, with implications for both academic research and practical applications in financial markets.

Key concepts

#claim/news_article; #news_article; #sentiment_analysis; #social_media; #finding/bidirectional_encoder_representations_from_transformers; #bidirectional_encoder_representations_from_transformers; #claim/dynamics; #dynamics; #claim/large_language_models; #large_language_models; #finding/GPT_4; #GPT_4; #claim/natural_language_processing; #natural_language_processing

Quote

The study explores the use of natural language processing (NLP) models, including BERT and transformer-based models, to analyze sentiment in cryptocurrency-related text data and predict cryptocurrency price movements, demonstrating the potential of sentiment analysis in financial forecasting.

Key points

- The volatility of cryptocurrency markets is often influenced by a myriad of factors, including news articles, social media discussions, crypto signals, and investor sentiment ^{1, 2, 3}
- This study initially focuses on utilizing the innovative GPT-4 large language models (LLMs), alongside a parallel comparison with bidirectional encoder representations from transformers (BERT) and FinBERT natural language processing (NLP) models
- Results In Section 3, we explore the methodological approach used to assess the predictive abilities of the GPT-4 LLM, BERT, and FinBERT NLP models in crypto sentiment analysis and classification, seeking to understand their underlying complexities
- The integration of LLMs and NLP models for cryptocurrency sentiment analysis represents a powerful toolset that enhances investment decision-making in the dynamic cryptocurrency market

- This study showcases the efficacy of state-of-the-art models like GPT-4 and BERT in accurately interpreting and categorizing sentiments extracted from cryptocurrency news articles
- It is important to highlight that achieving an 82.9% accuracy rate in predictive modeling is quite high, indicating that the model can be confidently applied to sentiment analysis and classification tasks
- By providing a deeper understanding of sentiment dynamics in cryptocurrency markets, this study facilitates a more informed and data-driven approach to cryptocurrency investment, enabling investors to make well-informed decisions based on the real-time sentiment analysis of news articles and other relevant sources

Summary

Introduction

Cryptocurrencies are becoming increasingly prominent in financial investments, with more investors diversifying their portfolios and individuals drawn to their ease of use and decentralized financial opportunities.

The volatility of cryptocurrency markets is often influenced by a myriad of factors, including news articles, social media discussions, crypto signals, and investor sentiment.

Understanding and accurately gauging sentiment in this rapidly evolving environment is essential for making informed investment decisions and managing risks effectively.

Sentiment Analysis

Sentiment analysis techniques using advanced natural language processing (NLP) models have gained traction as valuable tools for extracting sentiment from cryptocurrency-related textual data.

The primary objective is to extract and quantify sentiment—positive, negative, or neutral—expressed in these texts to understand prevailing market sentiment towards specific cryptocurrencies or the broader market.

NLP methods are essential for discerning the emotions and opinions conveyed through language, with techniques including sentiment lexicons, deep learning approaches, and lexicon-based sentiment analysis.

The studies utilized various sentiment analysis techniques, including linear discriminant analysis, BERT, and transformer-based models, to predict cryptocurrency price movements.

Gurrib et al introduced a machine learning approach combining linear discriminant analysis and sentiment analysis, while Ider et al employed BERT classifiers and weak supervision techniques to improve predictive power.

Kulakowski et al developed sentiment classification solutions tailored to cryptocurrency-related social media posts, and Nguyen Thanh et al explored ChatGPT for sentiment analysis of Twitter data.

NLP Models

The study explores the capabilities of large language models (LLMs) and NLP models for analyzing sentiment in cryptocurrency-related news articles, and fine-tunes state-of-the-art models such as GPT-4, BERT, and FinBERT for this task.

The study aims to evaluate the performance of LLMs and NLP models in cryptocurrency sentiment analysis through a comparative classification study, providing insights to inform investment strategies and risk management practices in cryptocurrency markets.

The study also investigates the impact of fine-tuning LLMs and NLP models for specific tasks, aiming to enhance understanding of sentiment dynamics in the cryptocurrency market.

Time-Series Analysis

Georgoula et al. used a combination of time-series analysis and sentiment analysis to examine the determinants of Bitcoin prices, finding significant correlations between the Twitter sentiment ratio, public interest, and Bitcoin prices.

Chalkiadakis et al. introduced a novel framework to evaluate multimodal statistical causality in cryptocurrency markets, shedding light on sentiment-driven market behavior.

Hybrid Models

Several studies adopted hybrid approaches to improve cryptocurrency price prediction, including Critien et al, who

employed neural network models to enhance prediction accuracy, and Girsang et al, who proposed a hybrid LSTM and GRU model integrating social network sentiment analysis.

Serafini et al. compared statistical and deep learning approaches, highlighting the importance of sentiment in predicting stock market performance.

Model Fine-Tuning

The GPT-4 model was fine-tuned using the official OpenAI API, with a multi-epoch training strategy to adapt the pre-trained model for sentiment analysis of news articles.

The fine-tuning process involved data preparation, training configuration, and an iterative training process that refined the model's understanding and capabilities.

The fine-tuned GPT-4 model achieved an accuracy rate of 86.7%, outperforming the base model.

Other models, such as BERT and FinBERT, were also fine-tuned with different optimizers, with FinBERT achieving 84.3% accuracy and BERT achieving 83.3%.

Model Evaluation

The performance of the models was evaluated using various metrics, including accuracy, precision, recall, and F1-score.

The fine-tuned GPT-4 model emerged as the top performer, achieving high accuracy and balanced precision and recall.

The evaluation results showed that the fine-tuned models outperformed the base models, highlighting the importance of fine-tuning for specific tasks.

The choice of model may vary depending on specific use cases or priorities, with other models offering advantages in certain aspects, such as recall for specific classes or precision in particular scenarios.

Results And Discussion

The results of the comparative analysis of the GPT-4, BERT, and FinBERT models were presented, with the fine-tuned GPT-4 achieving the highest accuracy.

The discussion highlighted the importance of model evaluation and the need for fine-tuning to improve accuracy in predicting neutral sentiment.

The results also showed that the choice of optimizer can affect model performance, with the Adam optimizer outperforming the AdamW optimizer in some cases.

Further optimizations can be pursued by adjusting parameters to enhance overall model performance.

Model Performance

The study found that fine-tuned models such as GPT-4, BERT, and FinBERT achieved high accuracy in sentiment analysis of cryptocurrency news articles, with GPT-4 achieving a maximum of 86.7%.

The ADAM optimizer achieved higher accuracy than the ADAMW optimizer.

The FinBERT model achieved slightly higher accuracy when trained with the ADAMW optimizer, reaching 83.6%.

Each model achieved higher accuracy for specific labels, with the fine-tuned GPT-4 model more accurate at predicting positive labels, while BERT and FinBERT excelled at predicting neutral labels.

Fine-Tuning

Fine-tuning LLMs and NLP models for specialized tasks within domains like finance and cryptocurrency is essential to optimize their effectiveness in practical applications.

The study showed that fine-tuning these models on large, representative datasets enables them to develop a deeper understanding of the nuances and patterns unique to sentiment analysis, leading to more accurate predictions.

The adaptation involves learning to recognize crypto-related terminology, grasp context, and extract relevant features from textual inputs.

Cost And Usability

The study highlighted the trade-off between the cost and usability of LLMs and NLP models.

While the fine-tuned GPT model showed slightly higher accuracy, BERT is an open-source model available for free use in a self-hosted environment.

In contrast, GPT-4 requires access through the official OpenAI API, which incurs a cost.

The choice between the two models depends on the stakeholder's priorities: the GPT API offers a more convenient and sustainable solution for those seeking accurate, immediate predictions, while self-hosted solutions with BERT or similar NLP models are more cost-effective for companies with an AI team.

Study subjects

1000 crypto news articles

- used and a set after fine-tuning for cryptocurrency news sentiment. Remarkably, analysis and the classification. This section of 1000 crypto news articles is for this investigation. models demonstrated almost flawless prediction accuracy, few-shot achieving unveils research discoveries and post-fine-tuning insights gained by the authors' learning, concerning the effect a maximum of Specifically, the fine-tuned

49 papers

- Previous Studies on Sentiment Analysis in Cryptocurrencies. Our literature review presents a comprehensive overview of 49 papers on the evolving role of sentiment analysis in understanding and predicting cryptocurrency market dynamics. The studies showcase a diverse array of methodologies, ranging from traditional sentiment analysis to cutting-edge deep learning models, that highlight the significant impact of sentiment-driven factors on cryptocurrency price movements and investor behavior

Data analysis

- #method/time_series_analysis
- #method/bert_model
- #method/finbert_model
- #method/nlp_model
- #method/gpt_model

Findings

- It is important to highlight that achieving an 82.9% accuracy rate in predictive modeling is quite high, indicating that the model can be confidently applied to [sentiment analysis](#) and classification tasks
- The fine-tuned [GPT-4](#) model emerges as the top performer, boasting an impressive accuracy rate of 86.7%, closely followed by FinBERT with 84.3%, and [BERT](#) with 83.3%
- The F1-score, striking a balance between precision and recall for each class, showcases the ft:gpt-4 model's superior performance with the highest F1-score for class 1. The F1-score, striking a balance between precision and recall for each class, showcases its balanced precision and recall for that class
- In Figure 1, both the base and fine-tuned [GPT-4](#) models demonstrate a ballabels, despite achieving 88% accuracy in predicting negative labels
- 88% accuracy in sentiment prediction unveils the research and insights gained by the authors concerning the to
- [ADAMW optimizer led to a slight decrease in accuracy, around 3.2% lower compared to the ADAM optimizer](#)
- It is imperative to highlight that even without fine-tuning, the [GPT-4](#) base model achieved 82.9% accuracy

Builds on previous research

- Our research findings confirm that both the LLMs and NLP models are highly accurate in their predictions, with

the fine-tuned GPT demonstrating significantly higher performance compared to its base version and the BERT model. The complete codebase used in this study, including dataset cleaning classes, fine-tuning procedures, and datasets, is openly available in a GitHub repository under the MIT open-source license ⁴.

Contributions

- In summary, while each model displays strengths in specific metrics, it fine-tuned its balanced precision and recall for that class.

Limitations

- The limitations of the study include the use of a limited dataset and the potential biases in the NLP models used. The study also acknowledges the complexity of sentiment analysis in the cryptocurrency domain and the potential impact of external factors on cryptocurrency prices.

Future work

- The future work includes exploring the use of other NLP models and techniques, such as multimodal sentiment analysis and graph-based methods, to improve the accuracy of cryptocurrency price predictions. The study also suggests the potential of using sentiment analysis in other financial applications, such as stock market prediction and risk management.

References

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